

Infosys_HotelRevAI – AI Driven Revenue Analysis for Hotels

Milestone 3: DAX and Business Metrics

Progress Report – Milestone 3

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1. Overview

In Milestone 3, an interactive Operational Insight Dashboard was developed using Power BI to analyze hotel booking data. The primary focus of this milestone was to create meaningful business metrics using DAX (Data Analysis Expressions) and convert raw data into actionable insights. The dashboard enables analysis of booking trends, revenue performance, seasonal occupancy, and customer behavior.

2. Use of DAX in the Dashboard

DAX (Data Analysis Expressions) was used to create dynamic measures that respond to filters and slicers in the dashboard.

The following key measures were implemented:

1. Total Bookings

Total Bookings = COUNT(Bookings[Booking_ID])

This measure calculates the total number of reservations.

Business Purpose:

Used to track overall demand and compare performance year-wise.

2. Total Revenue

Total Revenue = SUM(Bookings[Revenue])

This measure calculates total income generated from bookings.

Business Purpose:

Helps in evaluating overall financial performance.

3. Weekend Revenue

Weekend Revenue = CALCULATE(SUM(Bookings[Revenue]), Bookings[Day_Type] "Weekend")

This measure calculates revenue generated specifically during weekends

Business Purpose:

Identifies peak earning periods.

Observation:

Weekend revenue contributes nearly 74% of total revenue.

4. Weekday Revenue

Weekday Revenue =

CALCULATE(SUM(Bookings[Revenue]),Bookings[Day_Type] = "Weekday")

This measure calculates weekday revenue separately.

Business Purpose:

Helps in identifying low-performing periods and planning weekday promotions.

5. Year-wise Bookings

Bookings by Year = CALCULATE([Total Bookings], ALLEXCEPT(Bookings, Bookings[Year]))

Used to compare bookings across different years.

Business Purpose:

Supports growth analysis and year-over-year comparison.

6. Average Revenue per Booking

Average Revenue per Booking = DIVIDE([Total Revenue], [Total Bookings], 0)

This measure calculates the average income generated per booking.

Business Purpose:

Helps in understanding pricing effectiveness and revenue efficiency.

3. Business Metrics and Insights**3.1 Booking Performance**

Total bookings were analyzed across years.

Insight:

2019 recorded the highest booking volume.

Business Impact:

Helps in evaluating yearly growth and operational planning.

3.2 Revenue Distribution

Revenue was split into weekend and weekday categories.

Insight:

Weekend revenue dominates total earnings.

Business Impact:

Indicates heavy dependence on weekend customers; weekday strategies can improve balance.

3.3 Monthly Occupancy Trend

Bookings were analyzed month-wise to identify seasonal patterns.

Example DAX for monthly analysis:

Monthly Bookings =

CALCULATE([Total Bookings],ALLEXCEPT(Bookings, Bookings[Month]))

Insight:

Higher occupancy observed between Month 8 and Month 10.

Business Impact:

Supports seasonal pricing and demand forecasting.

3.4 Stay Duration Analysis

Stay duration was analyzed to understand customer behavior.

Total Stay Nights = SUM(Bookings[Stay_Duration])

Insight:

Longer stays contribute significantly to total revenue.

Business Impact:

Encourages development of extended stay offers.

3.5 Special Request Analysis

Bookings were compared based on the number of special requests.

Insight:

Bookings decrease as special requests increase.

Business Impact:

Operational efficiency can be improved by simplifying service processes.

4. Data Modeling Approach

- Established relationships between booking and date tables.
- Maintained one-to-many relationships.
- Used measures instead of calculated columns for better performance.
- Managed filter context carefully using CALCULATE and ALLEXCEPT functions.

5. Key Outcomes

- Identified peak revenue periods.
- Analyzed seasonal booking trends.
- Evaluated customer stay behavior.
- Measured performance using structured KPIs.
- Applied DAX to generate dynamic business insights.

6. Conclusion

Milestone 3 successfully demonstrates the implementation of DAX to create business metrics and analytical KPIs. The developed dashboard transforms booking data into structured insights that support informed decision-making in revenue planning, customer strategy, and operational management.